

MAE 184 Flight Simulation Techniques
Homework #4 Due FRI 06 JUN 2025

Please submit final answers using the formatted "Quiz" on the Canvas course website. This worksheet is for reference only.

Problem 4.2 requires the JSBSim program and the T38 aircraft model that has been modified for MAE 184 (i.e. not the version from the FlightGear website). The T38 model for MAE 184 is available on Canvas and in the simulation lab.

Problem 4.1 (2 pts) LIS (Slide 1)

Write a function that will take four corner points of a 2D data table and estimate the value for any point that lies inside of the corner points. Demonstrate your function by estimating the value of thrust [N] for a point at Mach 0.44 and altitude 900 m.

Mach	Altitude = 0 m	Altitude = 2000 m
0.4	83487 N $f(x_1, y_1)$	68952 N $f(x_1, y_2)$
0.6	86328 N $f(x_2, y_1)$	71665 N $f(x_2, y_2)$

Thrust (Mach = 0.44, altitude = 900 m) = 77502.93 [N]

Problem 4.2 (2 pts) *Run sim 2x (w/ & w/o SAS) → store under different names & plot the csv files together
↳ change mach & altitude in elevator xml file (the ic condition values) *adjust in python*

Create a time response plot to illustrate the dynamic effect of the T-38 pitch SAS modeled in JSBSim. The plot should show the pitch rate (q) response [y-axis, units: deg/sec] versus time [x-axis, units: sec]. The plot will show two responses, one with the pitch SAS enabled and a second with the pitch SAS disabled. Be sure to label each axis and use a legend to differentiate the two response curves. Submit your plot using an image format (BMP, PNG, JPG).

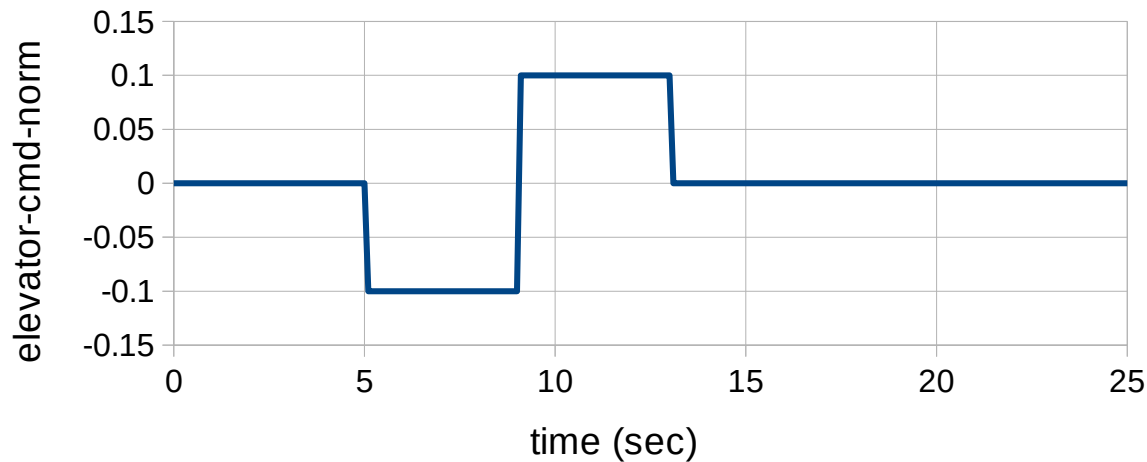
Aircraft: T38.xml (modified for MAE 184)

[Flight Condition: Mach = 0.8, altitude = 40,000 ft] ← adjust file input to this before running

To disable the T-38 pitch SAS, change the <gain> value in the aircraft file "T38.xml".

```
<!-- set gain to 0.0 to disable pitch SAS -->  
<gain>1.0</gain>
```

Input: Normalized Elevator Command (fcs/elevator-cmd-norm) as generated by script "T38_elevator.xml" and shown below.



Problem 4.3 (2 pts)

Estimate values for the two aerodynamic derivatives, C_{lp} and $C_{l\delta_A}$, from the flight test data provided on Canvas. Both parameters should be adjusted to yield the best match to flight test data, but only C_{lp} needs to be reported in the answer. **won't match exactly, but reasonably close*

Data File: T38_aileron.csv

↳ will require running code several times

[time(sec) aileron(rad) roll_rate(rad/s) roll_angle(rad)]

Flight Condition:

altitude, $h = 40,000$ ft; Mach, $M = 0.86$

dynamic pressure, $q_{bar} = 200$ lbf/ft²

airspeed, $V = 830$ ft/s

Aircraft Geometry:

wingspan, $b = 25.5$ ft; wing area, $S = 170$ ft²

moment of inertia, $I_{xx} = 7850$ slug/ft³

Equations of Motion:

$$I_{xx} \frac{dp}{dt} = \bar{q} S b C_l \quad \frac{d\phi}{dt} = p$$

$$C_l = C_{lp} \left(\frac{pb}{2V} \right) + C_{l\delta_A} \delta_A \leftarrow \begin{matrix} \text{may require reasonable} \\ \text{initial guess} \end{matrix}$$

p = roll rate [rad/s]

ϕ = roll attitude angle [rad]

δ_A = aileron [rad]

Answer: $C_{lp} = \underline{-0.2663}$

Problem 4.4 (2 pts) ← for loop until last equation of L23 (Slide 9) crosses 0 ← pick a range for theta
↳ may need slide 8 eqns

Find the two possible body positions, defined by symmetric positive and negative values of theta (tilt angle of leg L3), that result when solving the 'forward problem' for the 2D Planar Motion System described in Lecture 23 (Motion Systems).

$$p1 = p2 = p3 = \sqrt{5}$$

$$L1 = 2, L2 = L3 = \sqrt{2}$$

$$\gamma = \pi/2, x1 = 4, x2 = 0, y2 = 4$$

$$\text{Answer: } \theta = (+/-) \underline{45} \text{ [deg]}$$

Could do sys of eqn or
just plot last eqn of slide 9 (L23)

Problem 4.5 (2 pts) L24 (Slide 9)

Estimate the elevator hinge moment HM for a Cessna 182 flying at a level-flight trim condition where the dynamic pressure is 30 lb/ft^2 , angle-of-attack is $+1.8 \text{ deg}$, and the elevator is deflected $+2.0 \text{ deg}$.

$$Ch_0 = 0, Ch_{\alpha} = -0.06 \text{ (1/rad)}, Ch_{\delta E} = -0.6 \text{ (1/rad)} \quad \text{Convert deg} \rightarrow \text{rad}$$

$$\text{horiz. tail area} = 38.5 \text{ ft}^2$$

$$\text{horiz. tail chord} = 3.4 \text{ ft}$$

a. -3 ft.lb

c. -170 ft.lb

b. -90 ft.lb

d. -5140 ft.lb